

4.5.30. a. Use the quotient-remainder theorem with divisor equal to 3 to prove that the product of any two consecutive integers has the form $3k$ or $3k + 2$ for some integer k .

b. Use the mod notation to rewrite the result of

part (a).

a. Use the quotient-remainder theorem with

divisor equal to 3 to prove that the product of any two consecutive integers has the form $3k$ or $3k + 2$ for some integer k .

Suppose n is an integer.

Since two consecutive integers differ by 1, let the

two consecutive integers be n and $n + 1$.

We must show that the product of $n(n + 1)$ has the form $3k$ or $3k + 2$ for some integer k .

The quotient-remainder theorem:

" $n = dq + r$ and $0 \leq r < d$ "

Since the exercise wants us to use the divisor 3,

let $d = 3$

Then, $n = 3q + r$ and $0 \leq r < 3$

Since r has to be greater than or equal to 0, and

less than 3, we have three cases to test:

$r = 0$, $r = 1$, and $r = 2$

Remember, we are looking for the forms:

$3k$ or $3k + 2$, for some integer k .

Case 1: $r = 0$

$$n = 3q + 0$$

$$n(n + 1) = (3q)(3q + 1)$$

$$= 9q^2 + 3q$$

$$= 3(3q^2 + q)$$

$$k = 3q^2 + q, \text{ for some } k \in \mathbb{Z}$$

Therefore, $n(n + 1) = 3k$, when $r = 0$

Case 2: $r = 1$

$$n = 3q + 1$$

$$n(n + 1) = (3q + 1)(3q + 2)$$

$$= 9q^2 + 9q + 2$$

$$= 3(3q^2 + 3q) + 2$$

$$k = 3q^2 + 3q, \text{ for some } k \in \mathbb{Z}$$

Therefore, $n(n + 1) = 3k + 2$, when $r = 1$

Case 3: $r = 2$

$$n = 3q + 2$$

$$\begin{aligned}
n(n+1) &= (3q+2)(3q+3) \\
&= 9q^2 + 15q + 6 \\
&= 3(3q^2 + 5q + 2) \\
&= 3k, \text{ for some } k \in \mathbb{Z}
\end{aligned}$$

Therefore, $n(n+1) = 3k$, when $r = 2$

Solution: The product of two consecutive integers has the form $3k$ or $3k+2$ for some integer k , with divisor 3.

b. Use the mod notation to rewrite the result of

part (a).

Definition of mod:

$$n \bmod d = r \leftrightarrow n = dq + r$$

Solution: $n(n+1) \bmod 3 = 0$ or $n(n+1) \bmod 3 = 2$